



LAB ASSIGNMENT – 2

Complexity Comparison & Observed Growth

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2. Prepare a Complexity Comparison Table

The following comparison connects theoretical Big-O complexity with the practical growth trend seen in the benchmark runs.

Algorithm / Pattern	Best	Average	Worst	Space	Observed Growth
Linear Search	O(1)	O(n)	O(n)	O(1)	Increases with n
Binary Search	O(1)	O(log n)	O(log n)	O(1)	Very slow growth
Single Loop Traversal	O(n)	O(n)	O(n)	O(1)	Approximately linear
Nested Loop Traversal	O(n²)	O(n²)	O(n²)	O(1)	Rapid quadratic growth
Recursive Factorial	O(n)	O(n)	O(n)	O(n)	Linear time + stack growth
Iterative Factorial	O(n)	O(n)	O(n)	O(1)	Linear time, lower space

Theoretical vs Experimental Interpretation

- **Linear vs Binary Search:** Binary Search halves the search interval at every step, so its runtime grows much more slowly on sorted data.
- **Single vs Nested Loop:** A single loop performs one pass; the nested version repeats a full pass inside another, producing quadratic growth.
- **Recursive vs Iterative Factorial:** Both are O(n) in time, but recursion uses O(n) call-stack space while the iterative version uses O(1) auxiliary space.

Note: Runtime and memory values depend on the machine and measurement overhead. The complexity class is the stable theoretical result; measurements illustrate the trend.